Identifying Causes of Protectionism Using Double Machine Learning

Emily Harms · POL380 Paper Proposal

1 Proposed Topic

Why do some democracies protect domestic producers more than others? I aim to study whether factors such as exports, unemployment rates, GDP, inflation rates, and the number of political parties causally shape how protectionist a country's trade policy is. Using a panel of many countries spanning several decades, I will estimate the effect of each factor using double machine learning (DML), which lets flexible learners such as random forests and neural networks estimate causal parameters in the presence of complex confounding variables (Chernozhukov et al., 2016).

This project aims to answer the question **Which domestic factors cause countries to adopt more protectionist trade policies, and how robust are these effects?** by testing a variety of hypotheses regarding protectionism. Write a few sentences on why each is expected and the alternate is not expected

H1 (Exports): Exports being a higher percentage of GDP increases/decreases – which makes more sense? protectionism.

H2 (Umeployment): Increased unemployment rates cause increased protectionism.

H3 (GDP per Capita): Increases in GDP per capita causes increased protectionism

H4 (Total GDP in US$): A higher total GDP causes increased protectionism.

H5 (Inflation): Increased inflation rates cause increased protectionism.

H6 (Number of Political Parties): More political parties causes increased protectionism.

For the sake of this analysis, the level of protectionism will be quantified using tariff rates explain why this is a reasonable thing to do. In other words, higher tariff rates mean higher levels of protectionism, and vice versa.

2 Prior Work

3 Proposed Methods and Current Progress

This is summarizing the youtube series on DML (the green one that pops up when you google double machine learning youtube). Find the FWL Theorem paper and check this info against it, then cite FWL).

Double Machine Learning (DML) is based on the Frisch-Waugh-Lovell (FWL) Theorem, which says that interactions between variables can be combatted using multiple linear regression analyses. First, regress the treatment parameter D on the independent variable you'd like to test (X_T) to find the treatment residual (r_T). Then, regress the outcome (Y) on the nuisance variables (X_N) to obtain the nuisance residual (r_N), giving you the variation of the outcome independent of X_T. Finally, regress r_N on r_T. What does this last regression give you? DML takes the FWL Theorem a step further, expanding it beyond linear relationships. DML replaces linear regression with any machine learning algorithm I believe it's usually neural networks or random forest regressors – why these two and explain them. This enables the FWL process to be used on models in which the variables are not related linearly.

Add a paragraph on how prior analyses use linear regression. Mention the assumption about linear relationships and use as justification for DML. Also I don't think Ehrlich even attempted FWL, so that's not good either.

Data on tariff rates (World Bank Group, 2026*e*), exports (World Bank Group, 2026*a*), unemployment rates (World Bank Group, 2026*f*), GDP (World Bank Group, 2026*b*,*c*), and inflation (World Bank Group, 2026*d*) and the number of political parties (Varieties of Democracy, 2026) have been obtained, and the required data reduction and cleaning is complete. A simple Python program (Harms, 2026) was written to pull the data needed from each dataset, and add the data to one large .csv file. A subset of the data is shown in table 1. figure out how to to ref command so table numbers don't get mixed up

Country	Year	% Exports	Unemp.	GDP/Capita	GDP (US$)	Inflation	No. Parties	Tariff
ALB	1997	10.04	16.874	728.55	4.71e9	33.18	2.87	14.41
ALB	2000	19.53	19.023	1160.42	6.15e9	0.05	2.87	11.65
...	...	...	...	...	...	...	...	...
ZMB	2020	46.72	6.032	951.64	2.34e10	15.73	2.78	4.82

Table 1: Dataset that will be used for double machine learning. % Exports, unemployment rate, GDP/capita, GDP in US$, inflation rate, and number of political parties are independent variables, tariff rate is the dependent variable.

Put together a model (linear model, but just to show process) and go through FWL step by step to explain. Do this just for one variable and explain the for loop. Also I hate this gap between table and next paragraph, figure out how to get rid of it.

4 Impacts of Protectionism

5 Notes

- Check DML Paper citation – May be linking to ArXiv, find actual publication if so
- Is JeanReshef2017 peer reviewed? Double check.
- Check years for World Bank data and V-Dem data

Parameter	coef	std err	t	$P > \|t\|$	2.5 %	97.5 %
Exports	-0.0227	0.00962	-2.363	0.0181	-0.0416	-0.00388
Unemployment	0.0279	0.0511	0.546	0.585	-0.0722	0.128
GDP/Capita	-3.898	8.559e-6	-4.554	5.254e-6	-5.483e-5	-2.221e-5
GDP (US$)	-2.473e-13	0.00153	-1.610	0.107	-5.483e-13	5.375e-14
Inflation	-5.581e-4	0.00153	-0.366	0.715	-0.00355	0.00243
Number of Parties	0.0679	0.0725	0.936	0.350	-0.0743	0.210

Table 2: Preliminary results. A P-value less than 0.05 shows a statistically significant causal relationship.

References

Chernozhukov, Victor, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, and Whitney Newey. 2016. "Double Machine Learning for Treatment and Causal Parameters." : Forthcoming.

Harms, Emily. 2026. "Data Cleaning Code." https://github.com/harmse1tcnj2027-arch/POL380/blob/main/DataCleaning.py.

Varieties of Democracy. 2026. "Country-Party-Date: V-Party." Varieties of Democracy.

World Bank Group. 2026*a*. "Export of goods and services (% of GDP) [NE.EXP.GNFS.ZS]." World Development Indicators.

World Bank Group. 2026*b*. "GDP (constant 2015 US$) [NY.GDP.MKTP.KD]." World Development Indicators.

World Bank Group. 2026*c*. "GDP per capita (current US$) [NY.GDP.PCAP.CD]." World Development Indicators.

World Bank Group. 2026*d*. "Inflation, consumer prices (annual %) [FP.CPI.TOTL.ZG]." World Development Indicators.

World Bank Group. 2026*e*. "Tariff rate, applied, weighted mean, all products (%) [TM.TAX.MRCH.WM.AR.ZS]." World Development Indicators.

World Bank Group. 2026*f*. "Unemployment (% of total labor force) [SL.UEM.TOTL.ZS]." World Development Indicators.